Indian Institute of Technology Guwahati

Department of Physics

PH102

Tutorial-1

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1. Given a vector function $F = \hat{x}(3y - c_1 z) + \hat{y}(c_2 x - 2z) - \hat{z}(c_3 y + z)$
   (a) Determine the constants c_1, c_2 and c_3 if F is irrotational
   (b) Determine the scalar potential function V whose negative gradient equal F

2. Find the angle between the body diagonals of a cube

3. Use the cross product to find the components of the unit vector $\hat{n}$ perpendicular shaded plane in Figure 1

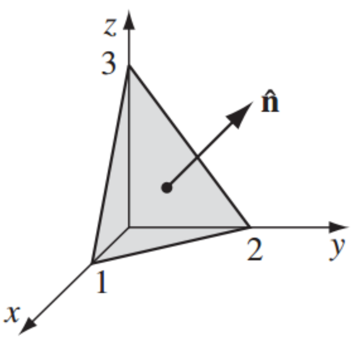

Figure 1

4. Let r be the separation vector from a fixed point (x', y', z') to the point (x,y,z), and let r be its length. Show that
   (a) $\nabla(r^2) = 2r$
   (b) $\nabla\left(\frac{1}{r}\right) = \frac{-\hat{r}}{r^2}$
   (c) What is the general formula for $\nabla(r^n)$

5. (a)Check product rule $\nabla \cdot (A \times B) = B \cdot (\nabla \times A) - A \cdot (\nabla \times B)$ for calculating each term separately for the functions

$$A = x\hat{x} + 2y\hat{y} + 3z\hat{z}$$
$$B = 3y\hat{x} - 2x\hat{y}$$

   (a) Verify the product rule $\nabla(A \cdot B) = A \times (\nabla \times B) + B \times (\nabla \times A) + (A \cdot \nabla)B + (B \cdot \nabla)A$
   (b) Verify the product rule

6. Derive the following quotient rules

$$\nabla \cdot \left(\frac{A}{g}\right) = \frac{g(\nabla \cdot A) - A \cdot (\nabla g)}{g^2}$$
$$\nabla \times \left(\frac{A}{g}\right) = \frac{g(\nabla \times A) + A \times (\nabla g)}{g^2}$$

7. Prove the following product rule

$$\text{(i)}\nabla(fg) = f\nabla g + g\nabla f$$
$$\text{(ii) } \nabla \cdot (A \times B) = B \cdot (\nabla \times A) - A \cdot (\nabla \times B)$$
$$\text{(iii) } \nabla \times (fA) = f(\nabla \times A) - A \times (\nabla f)$$